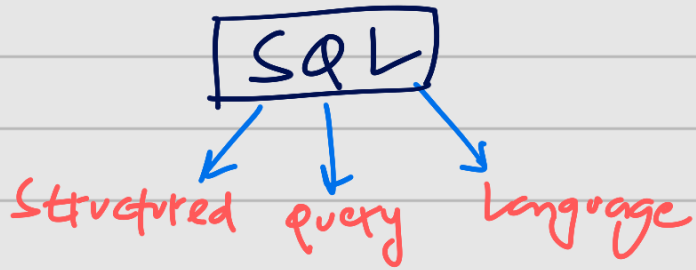


MySQL

Module 1 Intro to DBMS



⇒ DBMS?

- System that manages Database.
collection of data

⇒ why DBMS?

- To manage "Scaled data".
- Easy compared to conventional method (spread sheet)

⇒ Pros

- Deals with large dataset
- complex queries operation easily
- Real-time updates (Scheduled queries, by triggers)
- Efficient
- Fast
- less costly.

⇒ Database

collection of data (organized)

C (Create) → Add new tuple

R (Read) → search items

U (Update) → modify items

D (Delete) → Delete items

} 4 Core operations

→ Schema

- Structure of database
- Defines Tables, Columns, Relationships, constraints etc.

⇒ Keys in SQL

① Primary Key

- Unique identifier in each key.
- Can't have NULL values
- Must have Unique values
- Each table can have one only.

② Foreign Key

- Establishes relation b/w 2 tables.
- Refers to primary key of other table.
- Helps maintain referential integrity.

③ Unique Key

- Ensures all values in column are unique
- Can have NULL values
- Table can have multiple unique keys

④ Candidate Key

- Similar to Primary Key but multiple
- Primary Key is chosen from it.

→ DB Types

Relational

↳ Data is organized into tables (rows & columns)

Eg: MySQL

Non-Relational

↳ No specific schema

Eg: MongoDB

⇒ SQL

Language to interact with & manage database